

Milestone 3 | Frozen experimental protocol

Celiac duodenal endoscopy image-label reliability study | Version 1.0 | 20 September 2026 | Abheek Amin

Question and prespecified hypothesis

Question. Among images a held-out binary classifier predicts Normal, does a fixed review-priority score combining weak Normal confidence, instability under mild technical changes, and measured image quality identify more Celiac-labeled images than weak confidence alone when both flag the same number of images?

Directional hypothesis. At a review rate of 20% of predicted-Normal images, the combined score will flag more Celiac-labeled images than the confidence-only score. The unit is a distinct image; a flagged image is a label-prediction disagreement, not a verified missed diagnosis.

Data and inclusion decisions

Use the audited supplied archive (ZIP SHA-256 begins 60f3ab4a42b2): 188 readable JPEGs, including 89 Normal, 75 Celiac and 24 Doubtful. The binary candidate pool has 164 files. The dataset page says 192; the archive and article support 188, and the difference remains unexplained. The paper does not provide a filename-level inclusion list.

Remove repeated copies from the primary evaluation by keeping the lexicographically first filename in each exact byte-identical Normal group: n22/n64, n28/n61, n63/n67/n68 and n65/n66. This leaves 159 distinct images: 84 Normal and 75 Celiac. Retain all filenames, hashes, labels and group mappings in the manifest. No visually similar but nonidentical image is removed solely because of resemblance. Set all 24 Doubtful images aside from training and primary testing; their class name is not a quality judgment.

Variables and estimand

Inputs: Supplied Normal/Celiac image labels; one frozen classifier; its original and transformed predicted probabilities; focus, exposure and glare measurements on a central image region. The rank score is the experimental variable (confidence-only versus combined).

Target event: A held-out image with supplied label Celiac for which the classifier predicts Normal (probability of Normal at least 0.50). This is called a false normal relative to the supplied image label.

Primary outcome: Paired difference in the number of target events flagged by combined minus confidence-only ranking among the same held-out predicted-Normal set. Both flag exactly $K = \text{ceiling}(0.20 \times N)$ images, where N is that set size.

Scope. Labels were based on expert image interpretation. The supplied archive does not give patient IDs, biopsy results, serology, Marsh grade or per-image device mapping. No patient-level diagnostic or clinical-safety claim follows from this experiment.

Methods | Fixed before model development

All choices below are locked for the primary analysis. Record deviations with dates before looking at held-out outcomes.

Leakage control and held-out predictions

Create five stratified, group-disjoint folds using seed 20260920. Exact copies form one group before canonical deduplication. Keep the audit's strong visually related candidates together: c7/c8, c9/c10, c20/c21, c19/c6, b3/b7 and n54/n55. The c2/d11 pair crosses into the excluded Doubtful set. Freeze the remaining candidate-pair review and group assignments from the audit ledger before training or inspecting prediction errors. Grouping means apparent frame/encounter similarity, not verified patient identity. If actual patient IDs become available before training, replace these groups with patient groups and record the change as a new protocol version.

For each outer fold, train only on the other four folds. From these training groups reserve about 20% as a stratified, group-disjoint validation set for early stopping; never use outer-fold labels for model selection. Each distinct image receives exactly one prediction from a model that did not train or validate on its group. Pool the five held-out predictions for the one primary ranking comparison. The absence of patient IDs prevents a confirmed patient-disjoint test.

One fixed classifier

ImageNet-pretrained ResNet-18; input 224 x 224 RGB, bilinear resize, ImageNet channel normalization. Replace the final layer with a two-class head; fine-tune the final residual block and head, freeze earlier blocks. Use class-weighted cross-entropy (inverse class frequency within training fold), Adam learning rate 0.0001, batch size 16, at most 50 epochs, early stopping after eight validation epochs without lower loss, and training seed 20260920. No test-driven hyperparameter search. Preserve original image field in the primary run; keep the same trained model for both ranking methods.

Two prespecified review rankings

Confidence baseline: For every predicted-Normal image, lower original-image probability of Normal means higher priority. In a binary classifier, margin and entropy yield the same ordering here, so they are not extra baselines.

Prediction instability: Mean absolute change from original Normal probability under four deterministic transformations applied separately: brightness x 0.90, brightness x 1.10, JPEG recompression at quality 75 and at quality 90. The classifier is held fixed; larger change means higher priority.

Technical quality: Measure only the central 60% width x 60% height of the original image, away from most border and text overlays. Compute (1) $1 / (1 + \text{variance of the grayscale Laplacian})$ for blur, (2) fraction of grayscale pixels below 20 or above 235 for extreme exposure, and (3) fraction with all RGB channels above 245 for glare. Higher values mean poorer technical quality. Rank and average these three measurements within the predicted-Normal set. These are technical proxies, not clinical labels.

Analysis and interpretation | Prespecified

The 20% review rate is an experimental comparison setting, not a clinical recommendation.

Combined score. Within the pooled held-out predicted-Normal set, convert weak confidence, instability and poor quality to percentile ranks, with larger percentile = greater review priority. Average their three percentiles with equal weights. Both methods select their top K; resolve exact ties by filename ascending. Ranking uses images and predictions only, never held-out labels. No weights or cutoffs are fitted to test outcomes.

Primary comparison

Report N predicted Normal, F Celiac-labeled among them, K flagged, true event counts caught by each method, and the paired count difference. Also show capture among F (if $F > 0$), flagged-set event fraction, and alerts on Normal-labeled images. Name individual events and overlaps without exposing any identifiable patient details. A positive difference favors the hypothesis only for this dataset and trained classifier.

Additional and sensitivity analyses

Repeat the exact paired comparison at 10% and 30% using $K = \text{ceiling}(\text{rate} \times N)$, reporting actual rounded review percentages. For the 20% analysis, report confidence-plus-instability and confidence-plus-quality equal-rank ablations to see which added cue contributes; these are secondary. Tabulate model confusion counts, image-label sensitivity/specificity, and the fraction of Celiac-labeled images left in unflagged Normal predictions. Report a group-resampled percentile 95% interval for the paired count difference (2,000 bootstrap resamples of frozen image-group predictions, recomputing K and both rankings). This interval is conditional on trained models and does not measure uncertainty from new patients or training reruns.

Run one predefined shortcut check: repeat the five-fold pipeline using a central-field crop (retain central 80% width and 70% height, excluding outer UI/border areas) with the same architecture, folds and scoring rules. Compare direction of the primary difference and classification errors across the audited visual layout categories. The crop may remove useful mucosa; treat this as a sensitivity analysis, not an automatically superior preprocessing choice. Inspect quality-measure distributions by class and layout to identify whether quality is acting as a label or device proxy.

Interpretation rules and reporting

If F is zero, capture is undefined and the hypothesis cannot be tested. If F is 1-4, present only descriptive counts and explicitly call any apparent advantage too fragile for a superiority claim. Otherwise report the observed paired difference and conditional interval without changing the outcome, ranking, review rate or grouping. Discuss visually related frames that may still span unknown patients, possible label error, device/text shortcuts, small sample size, and the unresolved 192-versus-188 discrepancy. These results do not establish celiac diagnosis from endoscopy or replace serology, biopsy or specialist review.

Reference and provenance

Primary inputs: Celiac_Dataset_Image_Audit_Report.pdf (20 Sep 2026) and Milestone_1_Celiac_Endoscopy_Literature_Comparison.pdf (20 Sep 2026). Source dataset: [Mendeley Data v2](#), DOI [10.17632/t278gdxwzc.2](#). Associated study: [Hamza-Cherif et al. \(2026\)](#), [accessible abstract and data statement](#). Its reported 87.39% accuracy is not a directly comparable target after deduplication and grouped testing.